

Aaryan Jha

Software Engineer

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PROFESSIONAL SUMMARY

Detail-oriented **Software Engineer** specializing in **ASP.NET Core**, multi-tenant architecture, and secure enterprise systems. Proven experience in developing **scalable backend solutions**, implementing **REST APIs**, and applying **Clean Architecture** principles. Skilled in collaborating across international teams and delivering performance-optimized applications. Passionate about solving real-world business challenges through robust backend engineering.

EXPERIENCE

Software Engineer

Vertex Special Technology, Lalitpur, Nepal

July 2024 - Present • On-site

- Developed a **payment system** within the accounting module for seamless transactions.
- Engineered an **OCR-based bank statement processing pipeline**, integrating external APIs and transforming **JSON data into validated financial transactions**.
- Enabled financial reporting (**GL, Trial Balance, P&L, Balance Sheet**) with **export support in JSON, CSV, Excel, and PDF**.
- Implemented **multi-tenant architecture, 2FA with trusted devices**, and enhanced security via **authenticator apps and device fingerprinting**.
- Developed and deployed **.NET Core** applications using **Docker containers**.
- Integrated **ZKTeco and Startups biometric** attendance devices for **real-time punch-in/out** tracking.
- Engineered a dynamic **RBAC system** with **runtime role creation and fine-grained (module/action-level) permissions**, securing access **across 325+ APIs**.
- Designed and developed core **HRMS modules** including **attendance tracking, leave management, employee lifecycle management**, and **recruitment workflows**.
- Restructured API architecture to improve **scalability, modularity, and maintainability**.
- Developed **OCR processing APIs** with **AWS S3, SignalR notifications**, and **webhook-based** async workflows.
- Designed **version-controlled document system** supporting edits, reprocessing, and multi-session tracking.
- Built secure **API Gateway** and implemented **API key management using Argon2** for external AI integrations.
- Designed **idempotent webhook patterns** for reliable asynchronous processing updates.
- Collaborated with cross-functional teams and conducted **knowledge transfer sessions across global teams (US, Nepal, Pakistan)**.

- Prepared technical documentation including **SRS, ERD, sequence diagrams, and architecture designs**.
- Implemented automated testing in .NET using **xUnit, Moq, and Shouldly**, improving code quality and maintainability.
- **Took end-to-end ownership of backend development** for multiple projects, from design to deployment.

Tutor & Final Year Project Supervisor

Islington College Kathmandu, Nepal

Aug 2025 - Present • On-site

- Deliver lectures on application development using **.NET Core Web API, MAUI, and Blazor**.
- Mentor final-year students throughout the project lifecycle, ensuring timely progress and quality outcomes.
- Provide technical guidance, code reviews, and recommendations for improvements.
- Evaluate student projects through **assessments, documentation review, and viva examinations**.
- Conduct **project defense sessions**, assessing technical depth and implementation quality.

SKILLS

- **Frameworks & Libraries:** ASP.NET Core, .NET MVC, Entity Framework, LINQ, Flutter, EPPlus, SignalR
- **Databases:** MS SQL, PostgreSQL
- **Web & APIs:** REST APIs, JWT, 2FA, Device Fingerprinting, JSON, CSV, Excel
- **Tools & Platforms:** Git, GitHub, IIS Deployment, BitBucket, Docker, Redis, RabbitMQ, Json, Csv
- **Architecture & Patterns:** Clean Architecture, Bloc
- **Methodologies:** Agile, Scrum, Jira, Trello
- **Testing:** xUnit, Moq, Shouldly, EF Core InMemory
- **Other:** Stripe Integration, IoT Systems

EDUCATION

BSc (Hons) Computing 2022 – 2025

Itahari International College (London Metropolitan University), Nepal

- Graduated with **74% (Straight A)**
- **Triple A Scholarship** for excellence in Academics, Attendance, and Discipline

AWARDS & HONORS

- Selected for International Exposure Trip to Thailand (2024)
- 3rd Place, 30-Second Pitch Challenge, DesignThinkers Academy, Thailand
- Mentor at Hack4SafeFood Hackathon – Guided winning team (Team Syntax Error)
- Triple A Scholarship for academic excellence, discipline, and attendance (year: 2024, 2025, 2026)

LANGUAGES

• English (Fluent) • Nepali (Native) • Hindi (Conversational)

REFERENCES

• Nabaraj Ghimire (Principal Software Engineer, SELISE Group) • +977 9846817216